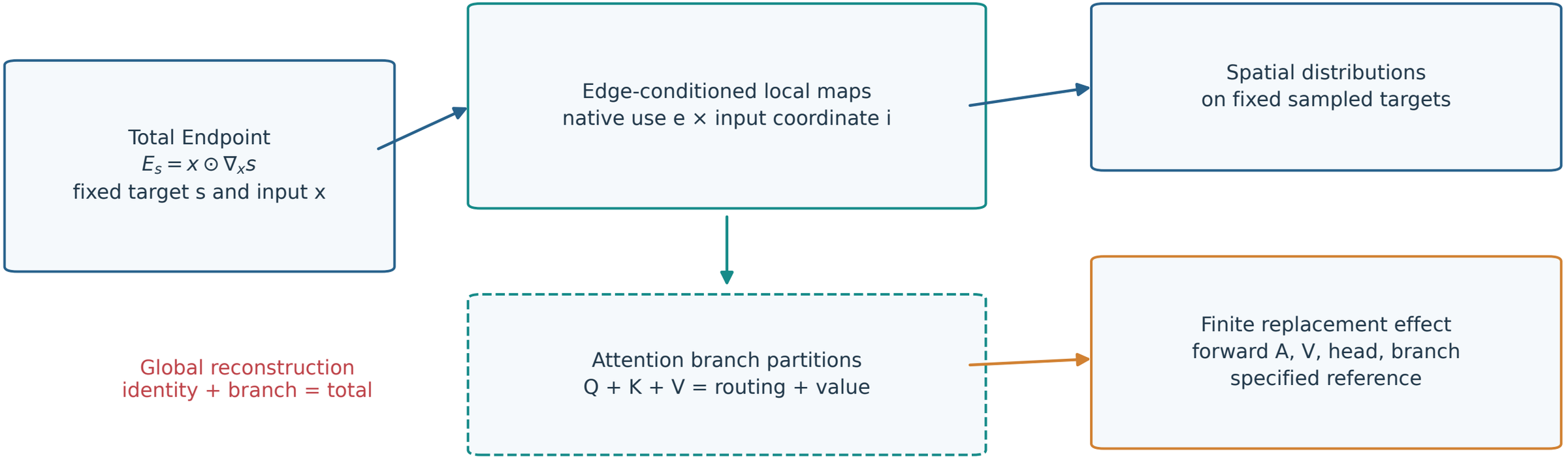Local differential maps: representation, reconstruction, and finite replacement